

2.A · Thermoelastic behaviour of condensed phases

2.A · Thermoelastic behaviour of condensed phases

Semester 1 · Academic Year 2026–2027

Family name:

Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The isothermal compressibility of water is $\chi_T = 4.5 \times 10^{-10} \text{ Pa}^{-1}$. Express it in bar^{-1} .

[illegible]

Marking

1	
0	1

Exercise 2. Thermoelastic coefficients — definitions

(5 marks)

Define the isothermal compressibility χ_T and the isobaric expansion coefficient α of a condensed phase. Give their SI units. Explain the minus sign in χ_T , and why both definitions divide by V .

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. Condensed phases

(3 marks)

For a liquid or a solid, the isothermal compressibility χ_T is:

- ☐ very small ($\chi_T \approx 0$) ☐ very large
☐ negative ☐ infinite

Expressed with the mass density ρ , χ_T equals:

- $$\begin{array}{ll} \square & \frac{1}{\rho} \left(\frac{\partial \rho}{\partial p} \right)_T \\ \square & \rho \left(\frac{\partial \rho}{\partial p} \right)_T \end{array} \qquad \begin{array}{ll} \square & -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial p} \right)_T \\ \square & -\frac{1}{V} \left(\frac{\partial \rho}{\partial p} \right)_T \end{array}$$

Over a restricted temperature range, the coefficient α of a condensed phase can be considered:

- ☐ constant
 - ☐ proportional to p
 - ☐ zero
 - ☐ negative

Exercise 4. Expansion and compression of water

(5 marks)

The coefficients of water are assumed constant. (a) A volume V_0 of water is heated at constant pressure from θ_1 to θ_2 . Estimate the change of volume ΔV . (b) Water is compressed at constant temperature from p_1 to p_2 . Estimate its relative change of volume $\Delta V/V$.

$$\alpha = 2.0 \times 10^{-4} \text{ K}^{-1} \quad \chi_T = 5.0 \times 10^{-10} \text{ Pa}^{-1} \quad V_0 = 250 \text{ mL}$$

$$\theta_1 = 20^\circ \text{C} \quad \theta_2 = 70^\circ \text{C} \quad p_1 = 1 \text{ bar} \quad p_2 = 101 \text{ bar}$$

A full-page view of a blank sheet of graph paper. The grid consists of small, uniform squares formed by thin gray lines. The paper has a white background and no margins or text are visible.

Marking

1

0	1
☐	☐

2

0	1
☐	☐

3

0	1
☐	☐

4

0	1
☐	☐

5

0	1
☐	☐

— End of the test —